Quick Start Guide on using GPU-As-A-Service Resource



Soh Hwee Jin, Melvin
Senior Assistant Director, InfoComm Infrastructure

Q1. What is GPU-As-A-Service (GaaS)?

GPU-As-A-Service (GaaS) is an new services offered by High Performance Computing Centre (HPCC) and consists of 25 Compute Nodes with 200 Nvidia H200 Cards.

To get the latest updates on what is happening to the Cluster, please subscribe to the NTU HPC Users' Group

Q2. How do I connect to GaaS Login Node?

Server IP Address: **172.21.33.7**

Step 1: Accessing Login Node with a SSH Client

- You can use MobaXterm (https://mobaxterm.mobatek.net/) to access the Cluster. Once you launch XTerm inside MobaXterm, you can use the follow command

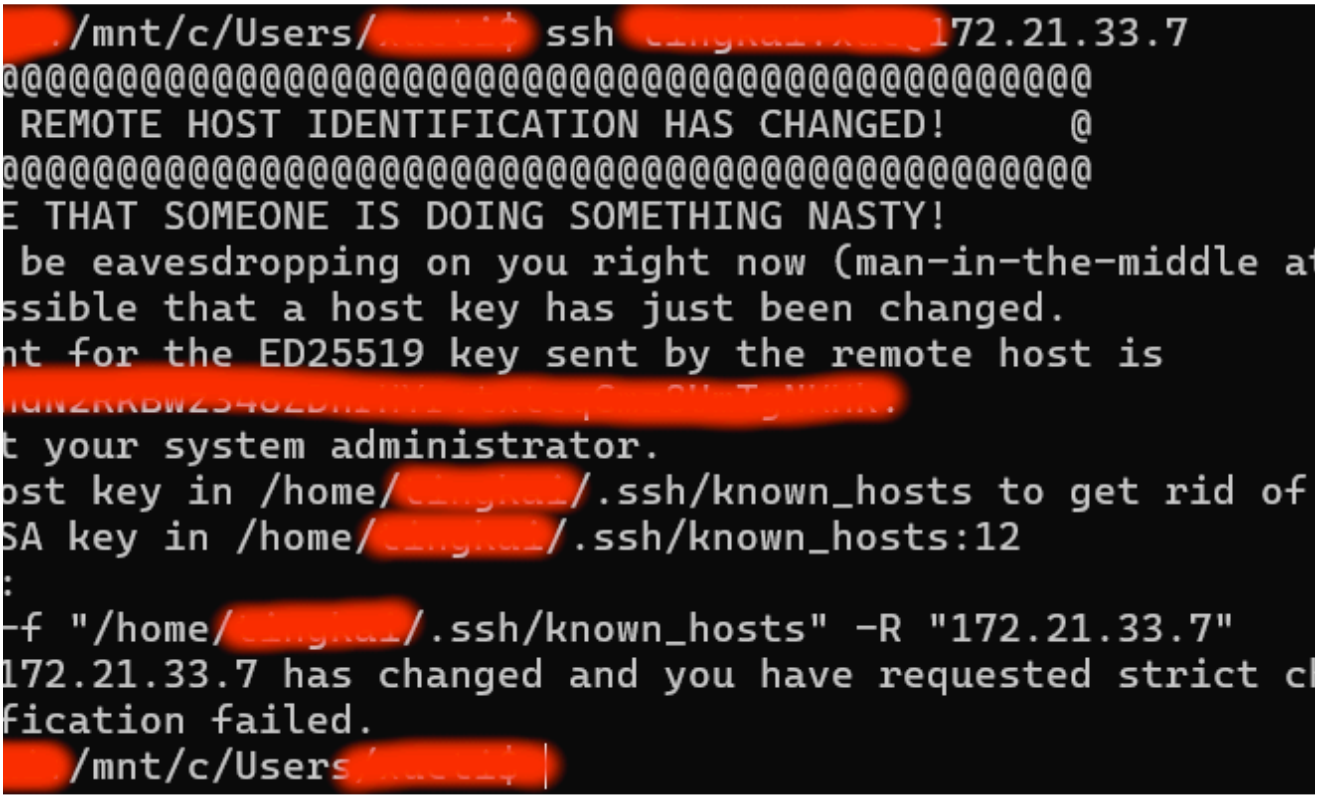
By DNS Name (Preferred)	By IP Address
$ ssh Your_NTU_UserID@gaas.hpc.ntu.edu.sg *Your_NTU_UserID@gaas.hpc.ntu.edu.sg* *Password: xxxxx*	*$ ssh Your_NTU_UserID@172.21.33.7* *Your_NTU_UserID@172.21.33.7* *Password: xxxxx*

- Alternatively, you can use Putty (https://www.putty.org/) to access the Cluster

- If you are working outside NTU, please remember to VPN in first. For students who are using the Cluster, do ask your supervisor to fill up a form to seek

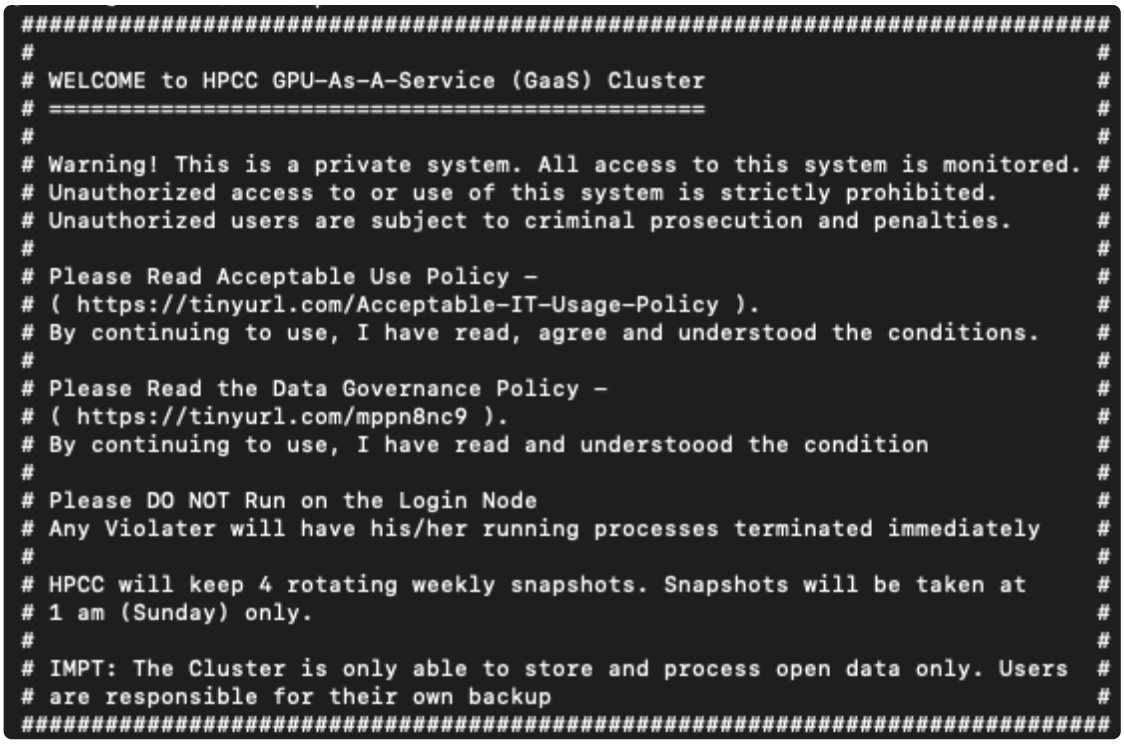
VPN access to Wildfly. For more information, do take a look at **Q20. How do I get VPN Access for Student Account?**

If you are facing errors such as the image below on your PC or MAC, you have to update the known_hosts on your client



S/No	Explnation	Command
	Remove a specific host's key from your known_hosts	ssh-keygen -R 172.21.33.7 ssh-keygen -R gaas.hpc.ntu.edu.sg

Upon successful login, you should see the Login Node. Kindly note the instructions written there.



Purpose of the Login Node:

- The Login Node is not a place for you to run your code. Please **DO NOT RUN** on the login node. The System Administrator will intervene to remove your processes as it interfere in other people work.
- There are more powerful nodes that can handle your codes are found in the compute resource pool. All you need is a submission script to get the resource to be assigned to you. "Q4 Where can I get sample Submission Script"

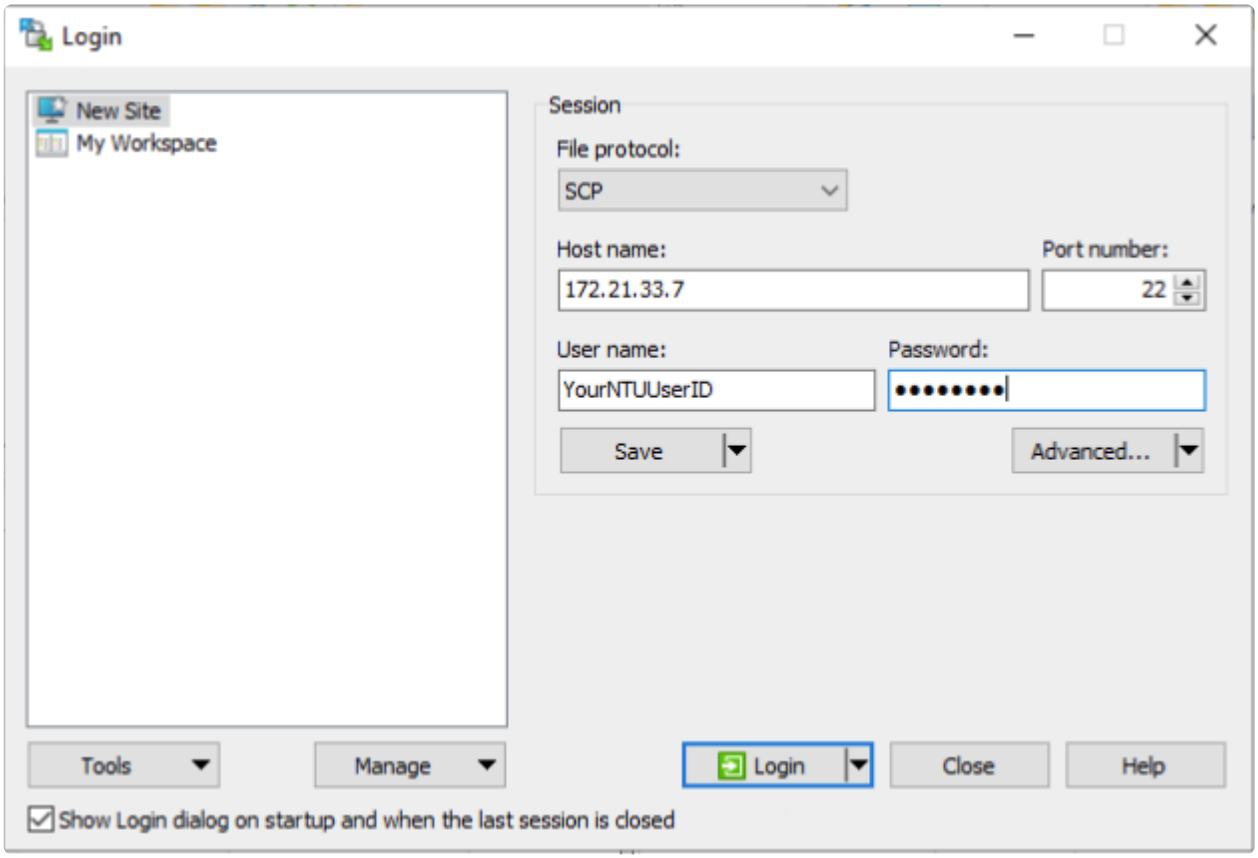
Note for Windows Users:

- If you are very uncomfortable using SSH Client, you may want to consider using WINSCP (https://winscp.net/eng/index.php) which is very Windows-like.
- Remember to use File Protocol: **SCP**

Further Readings on WINSCP

- Guides to Using WinSCP (By WinSCP)

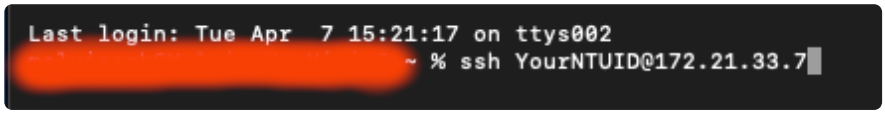
- WinSCP Tutorial - Connecting with FTP, FTPS, SFTP, uploading and downloading



Note for Mac Users:

Mac is UNIX under the hood. So all you need is to launch the MAC Terminal. User Your NTU ID (without Domain) and NTU Password

ssh Your_NTU_UserID@172.21.33.7
Your_NTU_UserID@172.21.33.7
Password: xxxxxxxx



If you wish to learn more about using MAC OS Terminal, you may wish to preview the Youtube Video by **Percy Grunwald** https://youtu.be/aKRYQsKR46I?si=ECG0O4oNNyvh9c3k

Q3. What applications are available on GaaS? How do I use them?

GaaS is using an environmental module system to load and unload libraries. It is not necessary to manually edit your own PATH and LD_LIBRARY_PATH. The understanding of this section will allows you to learn how to integrate the module load command in "Q4 Where can I get sample Submission Script"

Options	Explanation	Commands
1	To list the software installed at the cluster	*module avail*
2	To select the software paths and libraries	*module load anaconda/2025* *module load cuda/13.0*
3	To unload the software paths and libraries	module unload *anaconda/2025* module unload *cuda/13.0*
4	To list the software that you have loaded	module list

Q4. Where can I get sample Submission Scripts?

A High-Performance Computing Cluster has a large collection of powerful servers to process the job. When you submit a job, the Scheduler find the most suitable server according to your submission script parameters and place your job there.

Sample PBS Script can be found at /usr/local/templates/. Feel free to copy any scripts to your directory and modify them as you wish.

Steps	Options	Commands
1	List all Submission Scripts	*ls -l /usr/local/templates/*.pbs*
2	Copy the sample submission scripts into your directory	*cp /usr/local/templates/lammps.pbs .*

Q5. What do I edit in the Submission Script

Open up the submission script you have chosen (For example lammps.pbs) using your favourite editor. If you are using Linux, you can use favourite editors like vim or nano.

Step(s)	Explanation	Commands
1	Edit Job Name	*#PBS -N myMATLAB* (*Make sure there is no spaces in-between the letter)
2	Edit Project ID	*#PBS -P project_code* (*Check with your Supervisor on your specific Project Code. If you are using GPU Dedicated Subscription, there is a prefix ds_**** If you are using GPU Annual Subscription, there is a prefix gs_****)
3	Edit the Queue (Queue Represent the Resource Pool)	For Dedicated Queue *#PBS -q gpu_ded* For Annual Subscription Queue *#PBS -q gpu_as* For Pay-Per-Use *#PBS -q gpu_pps*
4	Edit the Walltime (Walltime represent how much time you are requesting from the scheduler)	*#PBS -l walltime=**120:00:00*** (Default is 120 hours)
6	Edit the Resource (GPU)	If I need 2 GPGPU (4 CPCPU Cores are attached to each GPGPU) *#PBS -l select=1:ncpus=8:ngpus=**2*** if I need 1 GPGPU (4 CPCPU Cores are attached to each GPGPU) *#PBS -l select=1:ncpus=**4**:ngpus=**1***
7	Edit the Resources (Mem)	I need a node with at least 1000GB RAM *#PBS -l select=1:ncpus=32:mem=1000g*
8	Load the Application you	*module load ansys/fluent/25R2*

Step(s)	Explanation	Commands
	need....	
9	For Anaconda Virtual Environment.....	Take a look at **Q8. How to compile and Test a GPGPU Application?**

Save and you are ready to submit your jobs. See Q6.

Q6. How do I submit my jobs?

You have to use SSH Client to login to GaaS to submit the job. See Q2 on how to Connect to GaaS. Once at the terminal, you can use the commands

S/No	Usage	Commands
1	General Commands	*qsub matlab.pbs*
2	View Job State on the Cluster	*qstat*
3	View my own jobs only	*qstat -u $USER*
4	View a specific Job Details	*qstat -f JobID* *(where JobID is the Job Number allocated)*
5	Kill a specific job that belongs to me	*qdel JobID* *(where JobID is the Job Number allocated)*
6	Check information on resources	*pbsnodes -aSj*
7	Checking Queue (Resources) Overview	*qstat -Q*
8	Launching an interactive mode (GPU)	*qsub -I -l select=1:ncpus=8:mpiprocs=8 -l walltime=04:00:00 -P my_projectid -q gpu_v100*

Q7. Can I have more information about the Queue?

Queue Name	Information	Resource Pool	Maximum Walltime per Job
gpu_free	Free Queue for PhD Students (* HPCC reserve the right	hpc-gaas-g25 (Total: 8 X H200)	4 hours

Queue Name	Information	Resource Pool	Maximum Walltime per Job
	to change allocation of resources to the queue)		
gpu_ded	Dedicated Queue for Projects	hpc-gaas-g01 to hpc-gaas-g05 (Total: 40 x H200)	1440 hours
gpu_gaas	Annual Subscription for Projects	hpc-gaas-06 to hpc-gaas-g24 (total: 152 X H200)	120 hours
gpu_ppu	Pay-Per-Use Usage	hpc-gaas-g06 to hpc-gaas-g24 (total: 152 X H200)	120 hours
gpu_datamover	Free for Users of GPU-As-A-Service	hpc-gaas-dm1 to hpc-gaas-dm2	120 hours

Q8. How to compile and Test a GPGPU Application?

The HPC Login Node is a lean Node with a L40s GPU. If you wish to compile based your application on the actual H200 card, do the following

Steps	Explanation	Commands
1	Request a GPU Resource through an Interactive Session	*qsub -I -l select=1:ncpus=4:ngpus=1 -l walltime=04:00:00 -P my_projectid -q gpu_as*
2	Load the Application of your course. For example, if you are using Anaconda Application Suite, load the ana	*module load anaconda/2025* *eval "$(/usr/local/anaconda2025/bin/conda shell.bash hook)"*
3	Build your virtual environment	See Set up virtual environment for Python using Anaconda
4	Make sure the GPU Submission Script have the following 3 lines:	## Load the Application *module load anaconda/2025*

Steps	Explanation	Commands
		eval "$(/usr/local/anaconda2025/bin/conda shell.bash hook)" *##(where myenv is the private virtual environment)* *conda activate myenv*

Q9. How do I run a multi-GPU / multi-node PyTorch job?

PyTorch distributed jobs on the cluster use NCCL over the GPU fabric, launched by mpirun from HPC-X (NVHPC 26.3). A single submission script (PYTORCH_MULTINODE_TEST.pbs) works for both single-node multi-GPU and multi-node runs — only the select= line changes.

Sample scripts are at /usr/local/templates/. Copy them to your directory and edit as needed.

Steps	Options	Commands
1	List all PyTorch sample scripts	ls -l /usr/local/templates/pytorch*
2	Copy the sample to your directory	cp /usr/local/templates/PYTORCH_MULTINODE_TEST.pbs /usr/local/templates/multinode_hello.py .
3	Single node, 8 GPUs — edit the resource line	#PBS -l select=1:ncpus=96:ngpus=8:mpiprocs=8:mem=2000gb
4	Multi-node (e.g. 2 × 8 GPUs) — edit the resource line	#PBS -l select=2:ncpus=96:ngpus=8:mpiprocs=8:mem=2000gb
5	Submit the job	qsub PYTORCH_MULTINODE_TEST.pbs
6	Check the output	cat PYTORCH_MULTINODE_TEST.o<JOBID>

Notes

- mpiprocs must equal ngpus (one rank per GPU).
- The queue gpu_as and project hpc_admin are required.
- The script uses pbs_tmrsh instead of SSH between nodes — no key setup needed.
- MASTER_ADDR / MASTER_PORT are set automatically from $PBS_NODEFILE; do not hardcode them.

Expected output of cat PYTORCH_MULTINODE_TEST.o<JOBID>

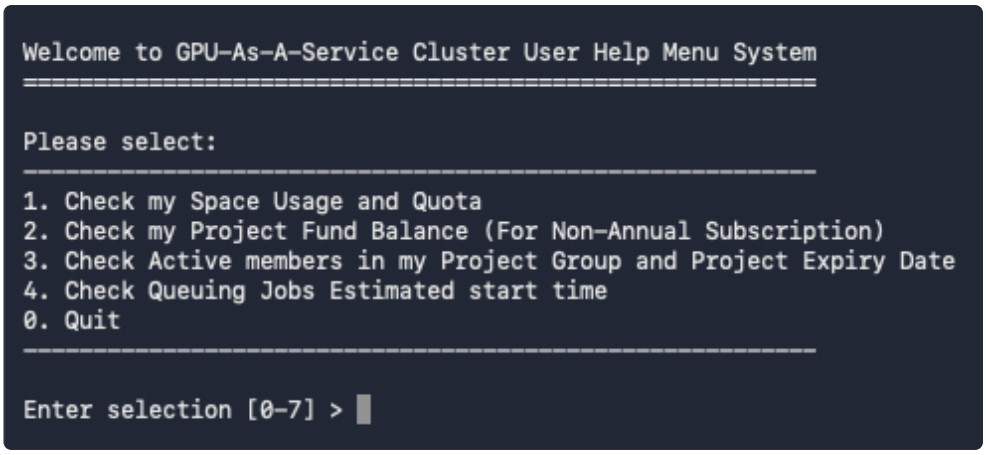
The output has three sections — a header banner, a per-rank GPU roll-call, and a pass/fail footer. Below is an example for 2 Node, 16 GPU test

```
Loading pytorch/2.11.0
 Loading requirement: gnu/gcc-12.3 python/3.12.13
============================================================
JOBID   : <JOBID>.gaas
WORKDIR : /home/<user>/<dir>
DATE    : <DATE> <TIME>
LAYOUT  : 2 nodes × 8 GPUs = 16 ranks
NODES   : hpc-gaas-gxx hpc-gaas-gxy
MASTER  : hpc-gaas-gxx:29500
MPIRUN  : mpirun (Open MPI) 4.1.9a1
============================================================
 rank  0/16 | hpc-gaas-gxx:GPU0 | NVIDIA H200
 rank  1/16 | hpc-gaas-gxx:GPU1 | NVIDIA H200
 ...
 rank 15/16 | hpc-gaas-gxy:GPU7 | NVIDIA H200

PASS — all_reduce + broadcast + matmul verified
 16 GPUs across 2 nodes
============================================================
==== TEST PASSED ====
```

Q10: What is GPU-As-A-Service Cluster User Help Menu System?

menu.sh is a Menu System to help Users with important information below



Screenshot of Menu.sh

S/No	Queries	Information Required	Commands
1	Check my Space Usage and Quota	You can quickly check your space usage and quota	menu.sh Select Option 1
2	Check my Project Fund Balance (For Non-Annual Subscription)	It will list the project groups that you are in, and the net balance of each project group. If you have an Annual Subscription and the number is unusually large or small, do not be alarmed. Annual Subscription is period-based, rather than funds-based. The funds inside the Annual Subscription are used for tracking usage rather than the actual net funds. In other words, any subscription which has gs_ If you are using Pay-Per-Use, the Net Balance in the Project Group is the Funds you have at HPCC	menu.sh Select Option 2
3	Check Active Group Members and Project Expiry in My Project Group	You can check who are the active members who has access to and can use your Supervisor Project Code. You can also check your Project Expiry Dates. This is useful if you need to verify the your CPU/GPU Annual subscription Project End Date.	menu.sh Select Option 3

S/No	Queries	Information Required	Commands
4	Check Queuing Job Estimated Start Time	Check the approximate time your queued jobs will run	menu.sh Select Option 4

Q11: How to use Data-movers Nodes to copy files to and from GPU-As-A-Cluster?

For more information see Using Datamover nodes to copy data to and from GPU-As-A-Service.